CLEANASAURUS · ALPHA

"STORAGE ALMOST FULL."

One developer ran a cleanup tool on their laptop and found **58.68 GB** in **128** old `node_modules` folders.

Most of them can be rebuilt with one command. Almost nobody deletes them, because deleting the **wrong** folder the night before a deadline costs you your project.

Source: Flavio Copes, ”npkill” (flaviocopes.com/npkill). Nate Kim · Nate Park · Wesley Chang

PENDING: confirm these numbers against Nate Kim's raw session evidence before presenting. Remove this banner when confirmed.

MVP VALIDATION · n = 10 (5 CS, 5 non-CS) · planned 15

Testers trusted the app. It didn't free the space they needed.

KPI	TARGET	RESULT	
Ran a plan in session 1 (trust)	≥ 60%	**70%**	MET
Unsafe actions reaching a tester (trust)	0	**0**	MET
Actions undone (trust)	≤ 10%	**11%**	CLOSE
Used again within 3 days (value)	≥ 40%	**30%**	CLOSE
Space freed vs. space wanted (value)	≥ 50%	**~2%**	MISSED

Outcome: PIVOT. Trust held; value failed. And our own bug: FREED showed 12.4 GB while **0 bytes** left the disk, because the Dino Bin was on the same drive.

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PIVOT: CS STUDENTS' REBUILDABLE DEV CLUTTER

38 GB

median rebuildable dev clutter per CS tester

1.4 GB

median duplicates our MVP could free

5 of 5 CS testers named a place the MVP wouldn't go.

The competition, honestly

CleanMyMac CLI (MacPaw, developer preview) already purges `node_modules` older than 7 days and clears Xcode DerivedData. npkill and DevCleaner each do one ecosystem.

What none of them do

- **Prove** a folder is rebuildable: lockfile, gitignored, no uncommitted work, no secrets
- Save the exact rebuild command
- Let Claude map "I'm done with CS 3140" to projects

LIVE DEMO · END TO END · 3½ MIN

One request, from a full disk to space back

1

Open a messy `~/code`: 8 projects, a VM, one with a `.env` secret

2

Type: "I'm done with CS 3140 and the hackathon. Free up space."

3

Claude picks the projects and writes the plan

4

Validator rejects 3 folders, with reasons. They go back to Claude, and try 2 passes

5

Approve: space freed, measured on disk. Undo shows the rebuild commands

Claude decides

Which projects "done with CS 3140" means · how to explain each action · when to ask instead of guess

Plain code decides

Whether a folder can be deleted (the rebuild proof) · what was really freed · nothing runs until you approve

Tricky data on purpose: uncommitted work, a project edited 4 days ago, a `.env` inside `node_modules`, a lockfile-less project, a VM disk. Backup: recorded run plus the offline planner.

BACKUP IF THE LIVE DEMO FAILS



Try 1: 6 folders proposed → 3 rejected (uncommitted work, active 4 days ago, a .env secret) → try 2 passed with 3 folders, each showing its rebuild command. ~150 MB measured freed. (Offline planner shown.)

METHODOLOGY & ENGINEERING METRICS

29
user stories

29 / 10
GitHub issues logged / closed
6 built in MVP, 4 by Alpha

114
automated tests
+ 14-step end-to-end UI test

3.9
tests per issue
every issue has ≥ 2

100%
pass rate of runnable tests (69/69)
45 written for unbuilt stories, kept red

94.5%
backend line coverage
88.6% branch

100% AI
tests generated by Claude · 10 refined after AI review · **human-refined: 0 so far**, team review before Beta

Tests first, no code (c1c337d) → Rebuild proof green (d2f885d, 454d384) → Claude-path tests red (3464586) → 69/69 green (c4c4aee)

Insight: the tests caught 2 real bugs (re-proposing source files as duplicates; ”6 versions of index.js”), and an independent review found 4 tests that couldn’t fail. All fixed.

STAGE & ROADMAP TO BETA

Alpha: done

- ✓ Validation run, decision made (Pivot)
- ✓ 29 stories → 29 GitHub issues with acceptance criteria
- ✓ Rebuild proof + JS and Python cleanup, real deletes
- ✓ Honest FREED (measured on disk)
- ✓ Claude path tested; CI on every push

Beta: 4 sprints, then a pivot test

1. "Done with CS 3140" via Claude (US-03) + storage breakdown
2. One-click rebuild, per-action undo, one-line summary
3. Xcode, Docker, ML caches · storage watch
4. Go Pro fake door · test kit

BETA PIVOT TEST: 20 CS STUDENTS WE DON'T KNOW	TARGET
First-session approval of a plan freeing ≥ 10 GB	≥ 60%
Median real space freed	≥ 10 GB
One-click rebuilds that work · unrecoverable deletions	≥ 95% · 0
Prefer us over CleanMyMac CLI purge, head to head (proposed)	≥ 50%
"Go Pro" clicks / pre-orders	≥ 15% / 5%

QUESTIONS?

Claude writes the plan. Plain code proves it’s safe. You get the space back.

github.com/natekim416/cs4501_startup · tests: `cd app && npm test`